SkyPredict AI — Viva Questions & Answers

Project-focused viva preparation based on the uploaded SkyPredict AI application.

Project snapshot

The application is described as an end-to-end Machine Learning application for flight-data exploration, ticket-price prediction and passenger-satisfaction analysis. It uses Random Forest Regressor for price prediction and Random Forest Classifier for satisfaction prediction. ■filecite markers are not reproduced in the PDF; the questions below are prepared directly from the project content.

1. What is SkyPredict AI?

Answer: SkyPredict AI is an end-to-end machine-learning application that explores flight data, predicts ticket prices, performs airline and route analytics, and predicts passenger satisfaction. The application is built with Python and Streamlit.

2. What are the main modules of the project?

Answer: The main modules are Flight Price Prediction, Dashboard/Analytics, Customer Satisfaction Prediction, and an About/Home section.

3. Which technologies are used?

Answer: Python, Pandas, NumPy, Streamlit, Scikit-Learn, Plotly and Joblib are used. The application also loads trained machine-learning models saved as .pkl files.

4. Why did you use Streamlit?

Answer: Streamlit allows a Python-based machine-learning project to be converted into an interactive web application quickly, without needing a separate frontend framework.

5. What is the objective of flight price prediction?

Answer: The objective is to estimate the expected ticket price from flight-related input features such as airline, route, travel timing, duration and stops.

6. Is flight price prediction classification or regression?

Answer: It is regression because the target, ticket price, is a continuous numerical value.

7. Which model is used for flight price prediction?

Answer: Random Forest Regressor is used in the application.

8. Why Random Forest Regressor?

Answer: Random Forest can model nonlinear relationships, handles interactions among features well, and is generally robust for tabular data. It combines multiple decision trees to improve generalization.

9. What is customer satisfaction prediction?

Answer: It predicts whether a passenger is satisfied or dissatisfied using passenger information and service-related features.

10. Is customer satisfaction prediction classification or regression?

Answer: It is classification because the target represents discrete classes such as satisfied and dissatisfied.

11. Which model is used for customer satisfaction?

Answer: Random Forest Classifier is used.

12. What is the difference between Random Forest Regressor and Classifier?

Answer: The regressor predicts a continuous numerical value, such as price. The classifier predicts a class label, such as satisfied or dissatisfied.

13. What is feature engineering?

Answer: Feature engineering means transforming raw data into useful model inputs. In this project it includes creating or converting features such as hour/minute components, duration and number of stops so that the model receives the expected schema.

14. Why is feature validation important?

Answer: The trained model expects a specific set of columns, names, data types and structure. If the application sends different columns, prediction can fail or become unreliable.

15. What is a machine-learning pipeline?

Answer: A pipeline chains preprocessing and modeling steps together so that the same transformations used during training are applied consistently during prediction.

16. Why use Joblib?

Answer: Joblib is used to save and load trained Scikit-Learn models efficiently. The Streamlit application loads the saved .pkl model instead of retraining it every time the app starts.

17. What is Pandas used for?

Answer: Pandas is used for reading CSV data, cleaning and transforming data, selecting columns, calculating statistics and preparing data for analysis and machine learning.

18. What is NumPy used for?

Answer: NumPy provides numerical arrays and mathematical operations and supports many data-science and machine-learning workflows.

19. What is Plotly used for?

Answer: Plotly is used to create interactive visualizations for airline comparisons, route analysis, stops, duration and trends.

20. What is the purpose of the dashboard?

Answer: The dashboard converts raw flight data into business-friendly insights such as airline performance, route patterns, stop distributions, duration analysis and monthly trends.

21. What is route validation?

Answer: Route validation checks whether the selected source and destination combination is available in the underlying flight dataset before attempting prediction.

22. Why should the model input match the training features?

Answer: A trained model learns relationships using the training feature schema. Supplying a different schema can cause errors or produce invalid predictions.

23. What happens when a user clicks Predict?

Answer: The application collects the inputs, validates them, performs the required feature engineering, creates a DataFrame with the expected columns, passes it to the trained model, and displays the predicted price and fare category.

24. Why is the predicted price prevented from becoming negative?

Answer: A negative ticket price is not logically meaningful, so the application applies a lower bound of zero to the model output.

25. What are the fare categories in the application?

Answer: The application categorizes predictions below ■4,000 as Budget Fare, ■4,000–■7,999 as Average Fare, and ■8,000 or above as Premium Fare.

26. What is overfitting?

Answer: Overfitting occurs when a model learns the training data too closely, including noise, and performs poorly on unseen data.

27. What is underfitting?

Answer: Underfitting occurs when a model is too simple to capture important patterns in the data, causing poor performance on both training and unseen data.

28. What is train-test split?

Answer: It separates data into training and testing subsets. The model learns from the training set and is evaluated on unseen testing data.

29. What is RMSE?

Answer: Root Mean Squared Error measures the typical magnitude of regression errors. Lower RMSE generally indicates better predictive performance.

30. What is accuracy?

Answer: Accuracy is the proportion of correctly classified observations out of all observations. It is one metric for evaluating the customer-satisfaction classifier.

31. What is a confusion matrix?

Answer: A confusion matrix summarizes classification results using true positives, true negatives, false positives and false negatives.

32. What is precision?

Answer: Precision is the proportion of predicted positive cases that are actually positive. It is TP divided by TP plus FP.

33. What is recall?

Answer: Recall is the proportion of actual positive cases correctly identified. It is TP divided by TP plus FN.

34. What is model serialization?

Answer: Model serialization means saving a trained model to a file so it can be loaded later without retraining. Joblib is used for this purpose in the project.

35. Why is the model not trained inside every Streamlit interaction?

Answer: Streamlit reruns the script when widgets change. Retraining on every interaction would be inefficient. Loading a pre-trained model makes prediction much faster.

36. What is session state in Streamlit?

Answer: Session state stores values across Streamlit reruns, such as the currently selected page or other user-session information.

37. Why are query parameters used in the app?

Answer: The application uses the URL query parameter to preserve the active page during Streamlit reruns and make navigation more reliable.

38. What is the role of st.rerun()?

Answer: It explicitly triggers a Streamlit script rerun after a navigation action so the application can render the newly selected page.

39. What is the biggest technical challenge in this project?

Answer: A major challenge is maintaining consistency between the training-time feature schema and the prediction-time input schema. Debugging model input columns, data types and preprocessing is therefore important.

40. How would you improve this project in the future?

Answer: I would add stronger model validation, cross-validation and hyperparameter tuning, explainability such as SHAP, monitoring for data drift, better handling of uncertainty, and deployment with authentication and production monitoring.

41. How would you explain this project to a non-technical interviewer?

Answer: I built an interactive analytics application that uses historical flight and passenger data to answer business questions: what a flight may cost, how airlines and routes behave, and what factors are associated with passenger satisfaction.

42. What business value does this project provide?

Answer: It demonstrates how historical data can support pricing estimation, route and airline analysis, and customer-experience insights. The broader value is converting data into actionable decisions rather than only building a model.

43. What would you say if asked whether the prediction is guaranteed?

Answer: No. A machine-learning prediction is an estimate based on historical patterns and available features. Actual prices can change because of demand, inventory, timing, seasonality and other factors not fully represented in the dataset.

44. Why is this an end-to-end project?

Answer: It covers the workflow from data exploration and preprocessing through model training, model serialization, application development, prediction, visualization and user-facing results.

45. Give a 30-second viva introduction.

Answer: SkyPredict AI is my end-to-end machine-learning project for flight analytics. It combines a flight-price regression model and a passenger-satisfaction classification model with an interactive Streamlit dashboard. I used Python, Pandas, Scikit-Learn, Plotly and Joblib to move from data preparation and model training to a usable prediction application.